Position, distance & displacement

Choose a reference point. Keep track of direction.

Name: _____ Date: _____

Show your steps and units for calculations. Use complete ideas for explanations.

Math warm-up • just the math
1. 12 + 5 = ____ **2.** 9 - (-6) = ____ **3.** 18 - 18 = ____

P01 The hallway delivery

CALCULATE

A student carries a box 12 m east from the classroom to the office, then walks 5 m west to a supply closet. What total distance does the student travel?

P02 Across the reference line

CALCULATE

On a straight east-west trail, the water fountain is position 0 m. A walker starts at -6 m and finishes at +9 m. Take east as positive. Find the displacement, including direction.

P03 One complete lap

EXPLAIN

A runner completes one full lap of a circular track and stops exactly where she started. Is her distance zero, her displacement zero, both, or neither? Explain using the start and finish.

P04 Seated on the train

EXPLAIN

A passenger stays seated while a train moves east at 10 m/s. Describe the passenger’s motion relative to the train, and then relative to the station platform. Why do the descriptions differ?

Explore these problems: https://gabrielw1005.github.io/physics-packet-lab/

Speed & velocity

Use a rate, a time, and a direction.

Name: _____ Date: _____

Show your steps and units for calculations. Use complete ideas for explanations.

Math warm-up • just the math
1. 180 ÷ 120 = ____ **2.** 4 × 45 = ____ **3.** Solve: 3t = 90. t = ____

P05 Walking the trail

CALCULATE

A student walks 180 m along a straight trail in 120 s. What is the student’s average speed? Include the correct unit.

P06 Cycling along the path

CALCULATE

A cyclist rides straight east at a steady 4 m/s for 45 s. How far does the cyclist travel?

P07 Jogging in opposite directions

EXPLAIN

Two joggers pass a fountain. One moves east at 2 m/s; the other moves west at 2 m/s. Do they have the same speed? Do they have the same velocity? Explain.

P08 The dog stops to sniff

EXPLAIN

A dog travels along a path at 2 m/s for 10 s, stops for 10 s, then continues at 2 m/s for 10 s. Is its average speed for all 30 s equal to 2 m/s? Explain why the stop matters.

Explore these problems: https://gabrielw1005.github.io/physics-packet-lab/

Acceleration

Compare velocities and use elapsed time. Right/east is positive.

Name: _____ Date: _____

Show your steps and units for calculations. Use complete ideas for explanations.

Math warm-up • just the math
1. 8 - 3 = ____ **2.** (14 - 4) ÷ 5 = ____ **3.** 6 + (-0.8 × 5) = ____

P09 The bus pulls away

CALCULATE

A bus travels east at 4 m/s when a stopwatch reads 3 s. At 8 s, it travels east at 14 m/s. What is its average acceleration during this interval?

P10 Braking on a scooter

CALCULATE

A scooter moves east at 6 m/s. The rider brakes with an average acceleration of -0.8 m/s² for 5 s, while still moving east. What is the final velocity?

P11 Backing up faster

EXPLAIN

A small remote-control car moves west. Its velocity changes from -1 m/s to -3 m/s in 2 s. Is it speeding up or slowing down? What does the negative sign tell you?

P12 Turning without speeding up

EXPLAIN

A cyclist follows a circular path at a steady 3 m/s. Is the cyclist accelerating even though the speed stays the same? Explain what is changing.

Explore these problems: https://gabrielw1005.github.io/physics-packet-lab/

Reading motion graphs

Read the axes first. Use a change in value divided by a change in time.

Name: _____ Date: _____

Show your steps and units for calculations. Use complete ideas for explanations.

Math warm-up • just the math
1. (30 - 0) ÷ (30 - 0) = ____ **2.** (8 - 2) ÷ (6 - 0) = ____ **3.** Compare: 8/10, 16/10, 4/10. Which is greatest?

P13 A pause on the nature walk

CALCULATE

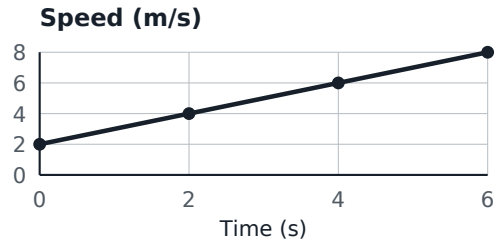
The table shows a student’s total distance along a trail. The student walks forward, pauses, then walks forward again. What is the average speed over the entire 30 s, including the pause?

Time (s)	0	10	20	30
Total distance (m)	0	12	12	30

P14 Reading a scooter’s speed graph

CALCULATE

The scooter’s speed increases steadily as shown. It travels east throughout. Use two points to find its acceleration from 0 s to 6 s.



P15 Which interval is fastest?

EXPLAIN

A walker’s distance-time data are shown below. Which 10 s interval has the greatest average speed? Explain how the distances, or the steepness of a graph, support your choice.

Time (s)	0	10	20	30
Total distance (m)	0	8	24	28

P16 What a speed graph cannot tell you

EXPLAIN

Two cyclists each ride steadily at 3 m/s for 6 s. One rides east and one rides west. Would their speed-time graphs differ? Could those graphs alone tell you their directions?

Explore these problems: https://gabrielw1005.github.io/physics-packet-lab/

Gravity & friction

Mass is measured in kg. Weight and friction are forces, measured in N.

Name: _____ Date: _____

Show your steps and units for calculations. Use complete ideas for explanations.

Math warm-up • just the math
1. 2.5 × 10 = ____ **2.** Solve: f - 35 = 0. f = ____ **3.** 1.6 × 150 = ____

P17 How heavy is the backpack?

CALCULATE

A backpack has a mass of 2.5 kg. What is its weight near Earth’s surface? Use g = 10 m/s².

P18 Keeping the box moving steadily

CALCULATE

A student pushes a plastic equipment box across a smooth floor at a steady speed. The push is 35 N to the right. What is the sliding friction force on the box? Give its size and direction.

P19 The same astronaut on the Moon

EXPLAIN

An astronaut and suit have a combined mass of 150 kg. Compare their mass and weight on Earth and the Moon, where gravity is weaker. Which changes, and why?

P20 A crate that will not move

EXPLAIN

A student pulls a crate right with 10 N, but it stays at rest. Static friction can reach a maximum of 18 N. What is the actual friction force now? Explain why it is not automatically 18 N.

Explore these problems: https://gabrielw1005.github.io/physics-packet-lab/

Net force & Newton’s first law

Combine forces on one object. Zero net force means zero acceleration.

Name: _____ Date: _____

Show your steps and units for calculations. Use complete ideas for explanations.

Math warm-up • just the math
1. 45 - 20 = ____ **2.** 18 + (-30) = ____ **3.** 18 - 18 = ____

P21 Pulling the supply cart

CALCULATE

A student pulls a supply cart right with 45 N. The total friction force is 20 N left. Find the horizontal net force on the cart.

P22 Two ropes, one cart

CALCULATE

Two students pull a low-friction cart in opposite directions. One pulls 18 N right; the other pulls 30 N left. Take right as positive. What is the net force? Give its sign and direction.

P23 Moving with balanced forces

EXPLAIN

A cart rolls right at a steady 1.2 m/s. The pull is 18 N right and friction is 18 N left. Must the cart stop because the forces balance? Explain what zero net force actually means.

P24 After the hockey stick lets go

EXPLAIN

A hockey puck glides across very smooth ice after the stick is no longer touching it. Over a short interval, ignore the small friction and air resistance. What happens to its horizontal velocity? Is a continuing forward force required?

Explore these problems: https://gabrielw1005.github.io/physics-packet-lab/

Newton’s second law

A net force changes velocity. More mass means more inertia.

Name: _____ Date: _____

Show your steps and units for calculations. Use complete ideas for explanations.

Math warm-up • just the math
1. 50 ÷ 25 = ____ **2.** Solve: 0.6m = 18. m = ____ **3.** 24 ÷ 12 = ___; 24 ÷ 24 = ____

P25 Accelerating the equipment cart

CALCULATE

A 25 kg equipment cart has a horizontal net force of 50 N to the right. What is its acceleration?

P26 Finding the trolley’s mass

CALCULATE

A loaded trolley accelerates right at 0.6 m/s² when its horizontal net force is 18 N right. What is the trolley’s mass, including its load?

P27 Empty cart or loaded cart?

EXPLAIN

The same 24 N net force acts on a 12 kg cart and a 24 kg cart. Both start at rest. Which accelerates more, and how does doubling the mass affect the acceleration?

P28 Force through a bicycle turn

EXPLAIN

A cyclist makes a circular turn on level pavement at constant speed. In which direction must the net horizontal force point: forward, outward, or toward the center? Which contact force from the road can provide it?

Explore these problems: https://gabrielw1005.github.io/physics-packet-lab/

Newton’s third law & momentum

A force pair acts on two objects. Momentum depends on mass and velocity.

Name: _____ Date: _____

Show your steps and units for calculations. Use complete ideas for explanations.

Math warm-up • just the math
1. 6 × 3 = ____ **2.** 0.5 × 1.2 = ____ **3.** 0.6 ÷ (0.5 + 0.5) = ____

P29 The bowling ball

CALCULATE

A 6 kg bowling ball rolls down the lane at 3 m/s. What is its forward momentum?

P30 Carts that latch together

CALCULATE

On a level, low-friction track, a 0.5 kg cart moving right at 1.2 m/s hits a stationary 0.5 kg cart. They latch together. Ignore external horizontal forces during the brief collision. What is their shared velocity immediately afterward?

P31 Skaters push apart

EXPLAIN

Two skaters, one 40 kg and one 60 kg, push off each other on smooth ice. During contact, does either exert a larger force on the other? Do equal force sizes mean equal accelerations? Explain.

P32 Pushing off the pool wall

EXPLAIN

A swimmer pushes backward on the pool wall. The wall pushes forward on the swimmer. Why do these equal and opposite forces not cancel when you find the net force on the swimmer?

Explore these problems: https://gabrielw1005.github.io/physics-packet-lab/